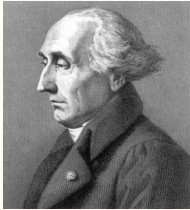


Optimization

Core ideas, one step at a time



Joseph-Louis Lagrange

Lagrange multipliers



William Karush

Inequality conditions, 1939



Harold Kuhn & Albert Tucker

Nonlinear programming, 1951

Part 1

Intuition

45 minutes: geometry, optimality and duality

Descartes: algebra meets geometry

Coordinates let an equation describe an entire shape.



René Descartes

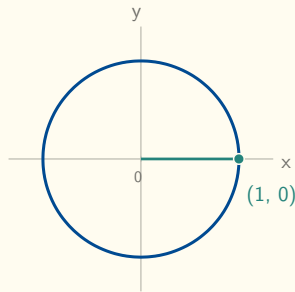
La Géométrie, 1637

"Cartesian" honours Descartes.

Algebra

$$x^2 + y^2 = 1 \longleftrightarrow$$

Geometry

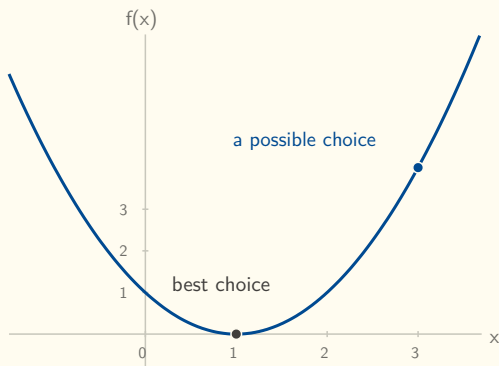


A circle of radius 1

We will describe the same optimization problem with formulas and geometric pictures.

Finding the minimum

Choosing variables to make an objective as small as possible

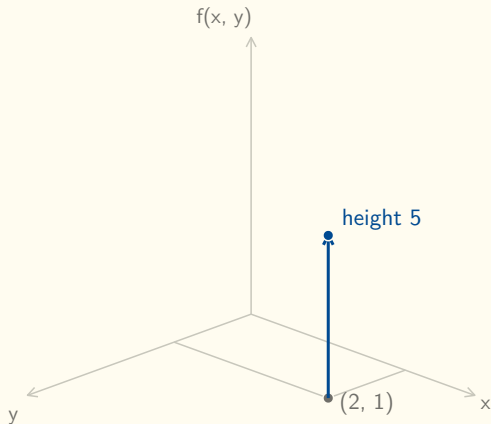


The variable is the choice.
The objective is the score.

$$\text{minimize } f(x) = (x - 1)^2$$

Two inputs, one height

A function turns an input pair into a score



$$f(x, y) = x^2 + y^2$$

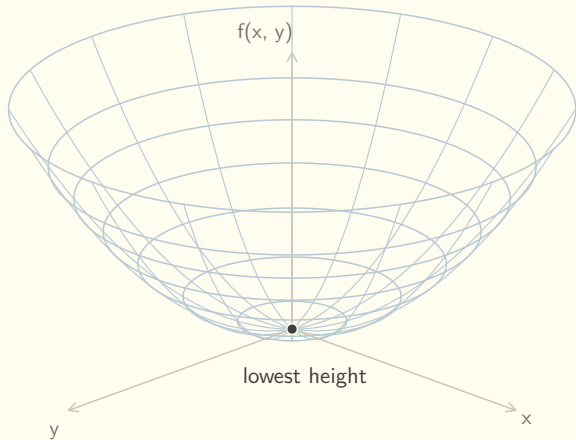
$$f(2, 1) = 2^2 + 1^2 = 5$$

Position: the inputs.

Height: the objective value.

A landscape of scores

All the heights together form a surface



$$f(x, y) = x^2 + y^2$$

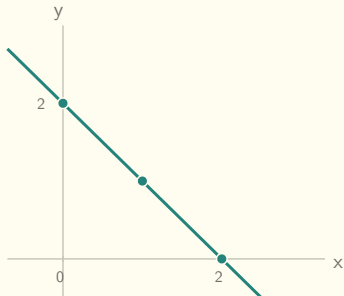
Higher means a larger score.
Lower means a smaller score.

Which input pair is lowest?

Constraints restrict movement

An equality acts like a rail. An inequality sets a boundary.

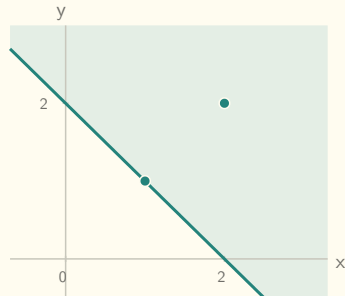
Equality



$$x + y = 2$$

$$h = 2 - x - y = 0$$

Inequality

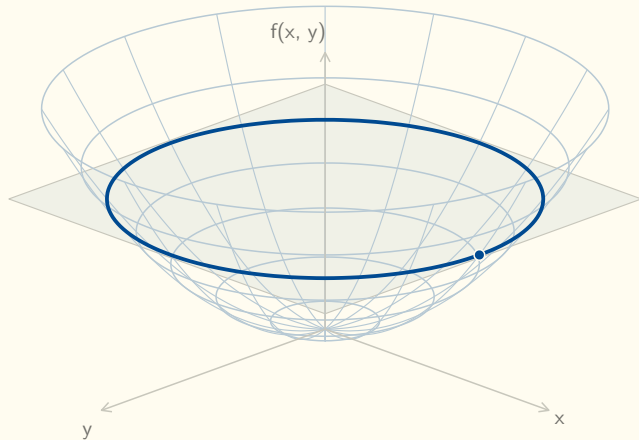


$$x + y \geq 2$$

$$g = 2 - x - y \leq 0$$

Points with the same score

Imagine a horizontal slice at height 4



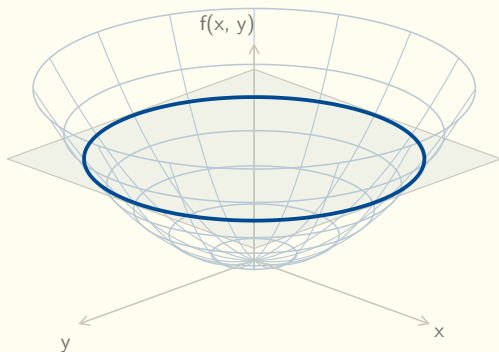
$$f(x, y) = 4$$

Every point on the blue ring has the same height.

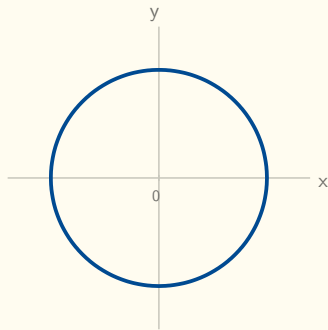
We keep only those points.

Looking from above

Keep the input positions of the points at height 4



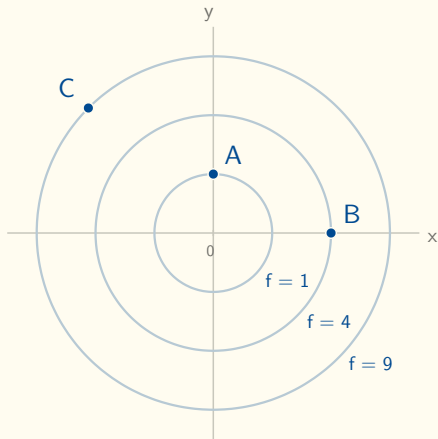
Surface: height = 4



Map: $x^2 + y^2 = 4$

A contour plot

Each curve joins input points with the same objective value



$$f(x, y) = x^2 + y^2$$

$$f(A) =$$

$$f(B) =$$

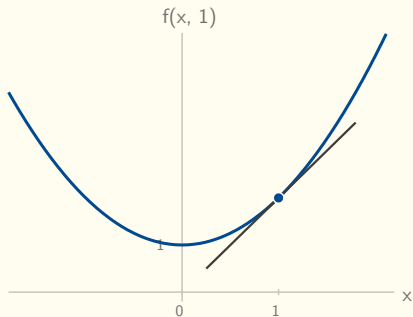
$$f(C) =$$

Which choice has the smallest score?

Change one input at a time

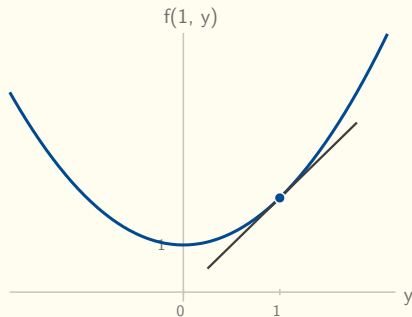
Each partial derivative measures one slope

Keep $y = 1$



$$\frac{\partial f}{\partial x} = 2x = 2 \quad \text{at } x = 1$$

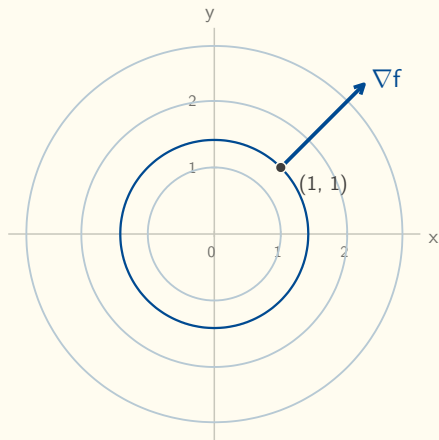
Keep $x = 1$



$$\frac{\partial f}{\partial y} = 2y = 2 \quad \text{at } y = 1$$

The gradient collects the slopes

Together, the slopes form a direction of steepest increase



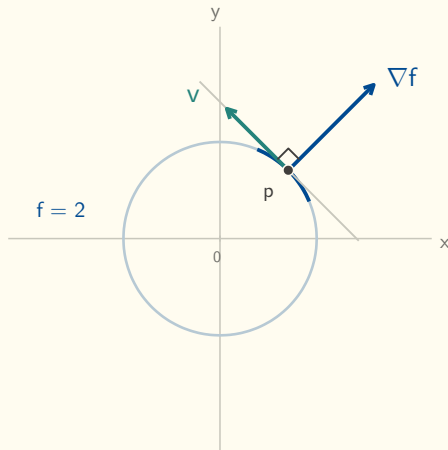
$$\nabla f = (\partial f / \partial x, \partial f / \partial y)$$

$$\nabla f(x, y) = (2x, 2y)$$

$$\text{At } (1, 1): \nabla f = (2, 2)$$

The tangent and the gradient

Along a contour, the score stays the same



v follows the curve locally.

$$v = (-1, 1)$$

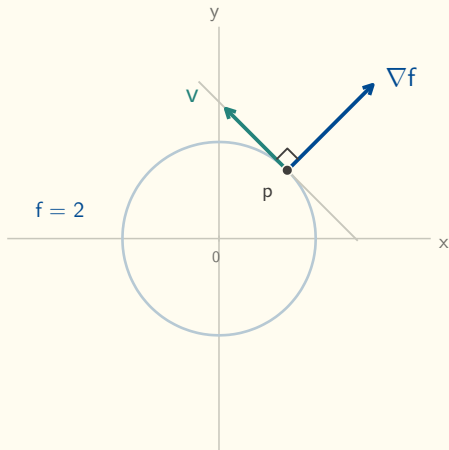
$$\nabla f = (2, 2)$$

$$\nabla f \cdot v = 2(-1) + 2(1) = 0$$

Zero dot product: a right angle.

Why the gradient is perpendicular

Let $(x(t), y(t))$ move along a contour



$$f(x(t), y(t)) = c$$

Differentiate along the curve:

$$0 = (\partial f / \partial x)x'(t) + (\partial f / \partial y)y'(t)$$

$$= \nabla f \cdot v$$

$v = (x'(t), y'(t))$ is the tangent.

At a smooth contour point with $\nabla f \neq 0$.

Recognizing the structure

The form of the problem helps us choose a method

minimize $x^2 + y^2$

———— Quadratic objective

subject to $x + y \geq 2$

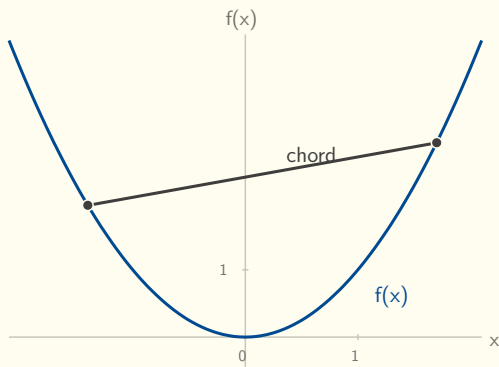
———— Linear constraint

Linear, quadratic, convex, nonlinear ...

A useful formulation makes useful structure visible.

Convex functions

Every chord lies on or above the graph

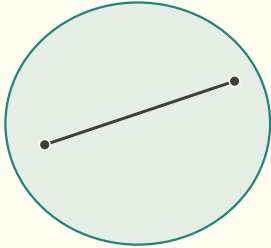


Mixing two choices never costs more than mixing their objective values.

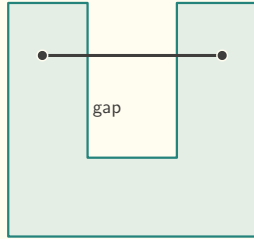
Straight lines and flat regions can be convex too.

Convex feasible sets

The segment between any two allowed points stays allowed



Convex

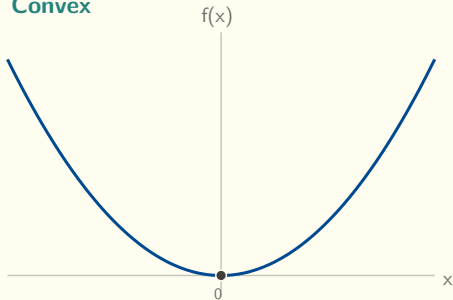


Not convex

Convex optimization

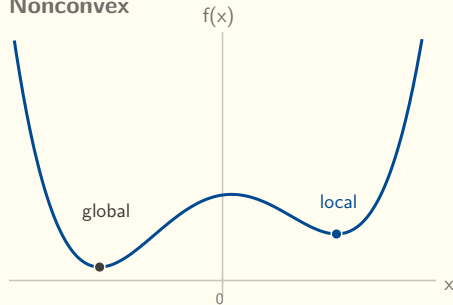
A convex objective over a convex feasible set

Convex



Every local minimum is global.

Nonconvex



A local minimum may not be best.

An equality as a min-max problem

The multiplier builds the equality into the objective

$$\min_z f(z)$$

subject to $h(z) = 0$



$$\min_z \max_{\lambda} L(z, \lambda)$$

λ can have either sign.

$$L(z, \lambda) = f(z) + \lambda h(z)$$

$$h = 0$$

$L = f$ for every λ .

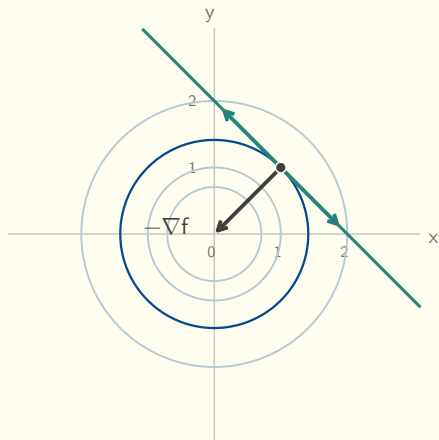
$$h \neq 0$$

Choose λ to make L grow without bound.

Min/max denote optimal values. Unbounded maximization is written $+\infty$.

First contact with the feasible set

The smallest reachable contour touches the equality



minimize $x^2 + y^2$

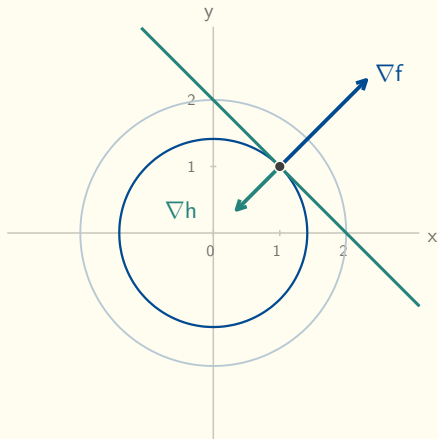
subject to $x + y = 2$

Downhill leaves the line.
Allowed moves stay on it.

No first-order decrease along the tangent.

At the best point, the curves touch

Both gradients are normal to the same tangent



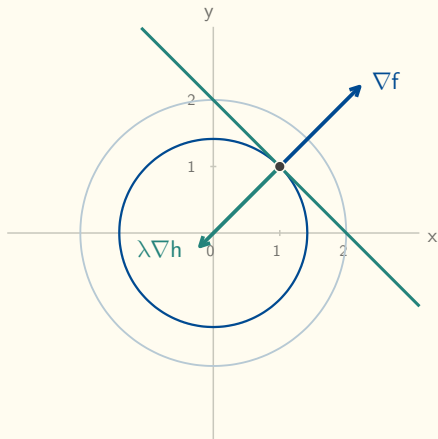
$$h(x, y) = 2 - x - y$$

$$\nabla h = (-1, -1)$$

The constraint gradient
is perpendicular to the line.

Lagrange multipliers

A multiplier scales the constraint gradient



$$\nabla f + \lambda \nabla h = 0$$

λ makes the vectors balance.

For an equality, λ can have either sign.

The Lagrangian: candidate equations

At a regular optimum, stationarity and feasibility hold together

$$L(z, \lambda) = f(z) + \lambda h(z)$$

Decision variables

$$\nabla_z L = \nabla f + \lambda \nabla h = 0$$

Equality multiplier

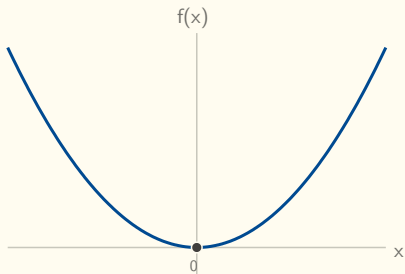
$$\partial L / \partial \lambda = h(z) = 0$$

Together: $\nabla_{(z, \lambda)} L = 0$

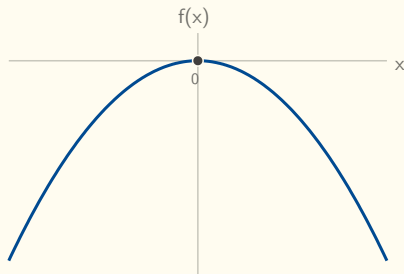
Then verify the candidate.

Stationarity gives candidates

A zero slope alone does not tell us whether a point is a minimum



x^2 : minimum at 0



$-x^2$: maximum at 0

Convex f + affine equalities: a feasible stationary point is a global minimum.

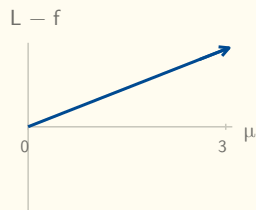
An inequality as a min-max problem

Minimize $f(z)$, subject to $g(z) \leq 0$

$$L(z, \mu) = f(z) + \mu g(z)$$

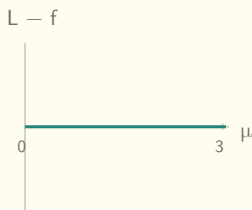
$$\min_z \max_{\mu \geq 0} L(z, \mu)$$

Violated: $g > 0$



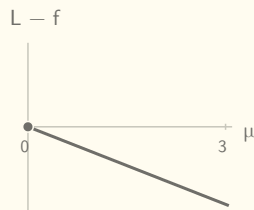
$$\max_{\mu \geq 0} L = +\infty$$

Boundary: $g = 0$



$$\max_{\mu \geq 0} L = f(z)$$

Interior: $g < 0$



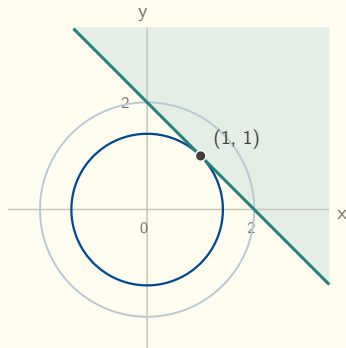
$$\max_{\mu \geq 0} L = f(z)$$

The inner maximum rules out every infeasible point.

An inequality can be active or inactive

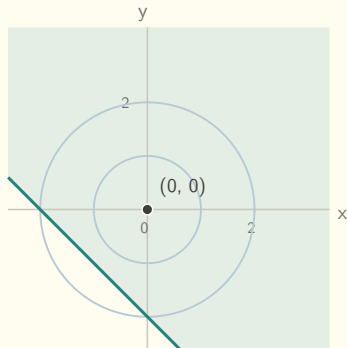
A constraint may touch the solution, or leave room

Active at the solution



$$x + y \geq 2$$

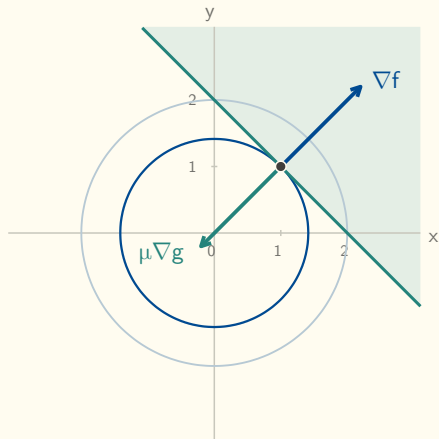
Inactive at the solution



$$x + y \geq -2$$

The inequality fixes the sign

With $g \leq 0$, the multiplier satisfies $\mu \geq 0$



$$\nabla f + \mu \nabla g = 0$$

$$\mu \geq 0$$

The sign follows our convention $g \leq 0$.

Complementary slackness

An inactive inequality has zero multiplier

$$\mu g = 0$$

Inactive

$$g < 0$$

$$\mu = 0$$

Active

$$g = 0$$

$$\mu \geq 0$$

An active constraint can still have $\mu = 0$.

The KKT conditions

Karush–Kuhn–Tucker conditions for smooth constrained optimization

$$L = f + \lambda h + \mu g$$

Allowed choice

$$h = 0, \quad g \leq 0$$

Balanced gradients

$$\nabla f + \lambda \nabla h + \mu \nabla g = 0$$

Nonnegative inequality multiplier

$$\mu \geq 0$$

Complementary slackness

$$\mu g = 0$$

One equality and one inequality shown. λ is unrestricted.

When KKT proves optimality

Convexity turns the conditions into a global certificate

For a differentiable convex problem

KKT satisfied $\longrightarrow$ **Global optimum**

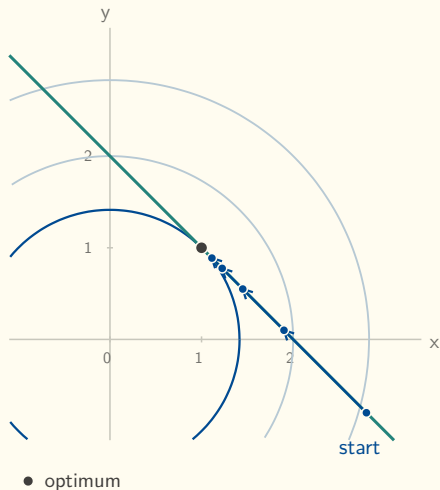
f and g are convex. Equality constraints are affine.

For necessity: an optimum satisfies KKT under a regularity condition.

Slater's condition is one useful condition for convex problems.

How solvers use these ideas

A numerical method approaches an optimum through successive choices



Use the problem's structure to choose useful steps.

Check feasibility and optimality conditions.

Illustration: feasible iterates on $x + y = 2$.

Reverse the order

The dual chooses a multiplier, then minimizes over the point

Primal

$$p^* = \min_z \max_{\mu \geq 0} L(z, \mu)$$

Choose z first.

Then let μ respond.

Dual

$$d^* = \max_{\mu \geq 0} \min_z L(z, \mu)$$

Choose μ first.

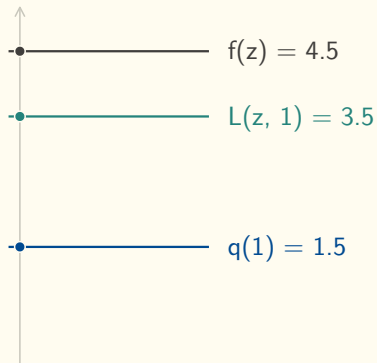
Then let z respond.

Useful with few constraints, or when the inner minimization separates.

$$q(\mu) = \min_z L(z, \mu)$$

Every multiplier gives a lower bound

For any feasible point z and any $\mu \geq 0$



$\mu = 1, \quad z = (1.5, 1.5)$

Minimize over every point:

$$q(\mu) \leq L(z, \mu)$$

At a feasible point, $\mu g(z) \leq 0$:

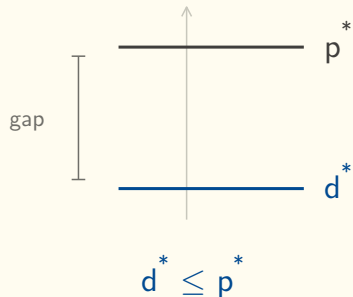
$$L(z, \mu) \leq f(z)$$

So $q(\mu) \leq p^*$

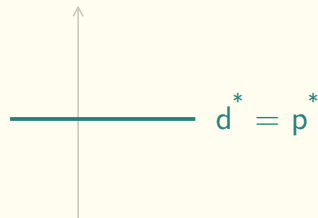
Weak and strong duality

Can the best lower bound reach the primal value?

Weak duality: always



Strong duality: zero gap



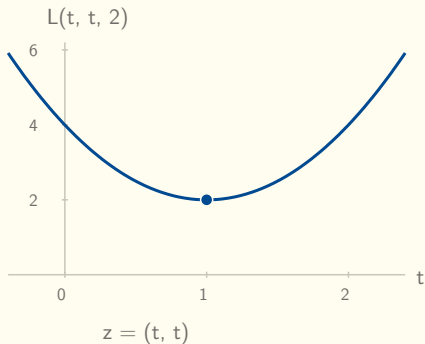
Convexity + a strictly feasible point

Slater's condition gives a standard route to strong duality.

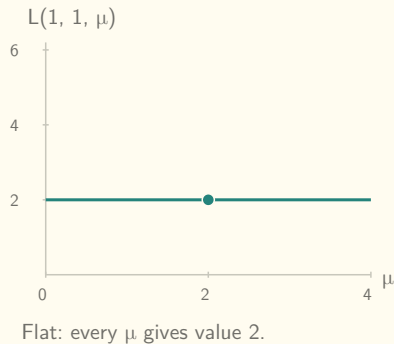
At an attained primal-dual solution

For a convex problem, KKT describes this saddle point

Fix $\mu = 2$. Move the point.



Fix $z = (1, 1)$. Move μ .



$$L(z^*, \mu) \leq L(z^*, \mu^*) \leq L(z, \mu^*)$$

Lecture break

Next: worked examples

Part 2

Examples

45 minutes: recipes followed by worked examples

Equality constraints: the recipe

Minimize $f(z)$, subject to $h(z) = 0$

1. Form the min-max problem

$$\min_z \max_{\lambda} L(z, \lambda)$$

$$L = f(z) + \lambda h(z)$$

2. Solve together

$$\nabla f + \lambda \nabla h = 0$$

$$h(z) = 0$$

3. Prove it is a minimum

Show no feasible point does better.

For example: a lower bound or convexity.

One multiplier per equality. λ can have either sign.

Smooth functions and regular constraints assumed.

Inequality constraints: the recipe

Minimize $f(z)$, subject to $g(z) \leq 0$

1. Standardize the constraint

$$g(z) \leq 0$$

2. Form the min-max problem

$$\min_z \max_{\mu \geq 0} L(z, \mu)$$

$$L = f(z) + \mu g(z)$$

3. Find KKT candidates

$$\nabla f + \mu \nabla g = 0$$

$$g(z) \leq 0, \quad \mu \geq 0$$

$$\mu g(z) = 0$$

For convex f and g , these checks prove global optimality.

Inequality recipe: the two cases

Complementary slackness tells us what to try

$$\mu g(z) = 0$$


A diagram with two arrows originating from the equation $\mu g(z) = 0$. One arrow points down and to the left towards the text 'Try $\mu = 0$ '. The other arrow points down and to the right towards the text 'Try $g(z) = 0$ '.

Try $\mu = 0$

Solve $\nabla f = 0$.

Check $g(z) \leq 0$.

Try $g(z) = 0$

Solve $\nabla f + \mu \nabla g = 0$
together with $g(z) = 0$.

Check $\mu \geq 0$.

The cases can overlap: $g = 0$ and $\mu = 0$ are both possible.

Several inequalities: try active sets, then check every constraint.

Duality: the recipe

Minimize $f(z)$, subject to $g(z) \leq 0$

$$L(z, \mu) = f(z) + \mu g(z) \quad \mu \geq 0$$

1. Fix μ . Minimize over z .

$$q(\mu) = \min_z L(z, \mu)$$

2. Maximize the lower bound.

$$\max_{\mu \geq 0} q(\mu)$$

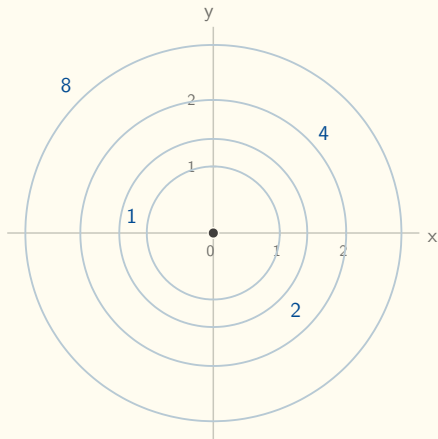
3. Compare with a feasible point.

$$q(\mu) \leq p^* \leq f(z)$$

Equal bounds certify the optimum.

Equality example: objective contours

Minimize $x^2 + y^2$ subject to $x + y = 2$



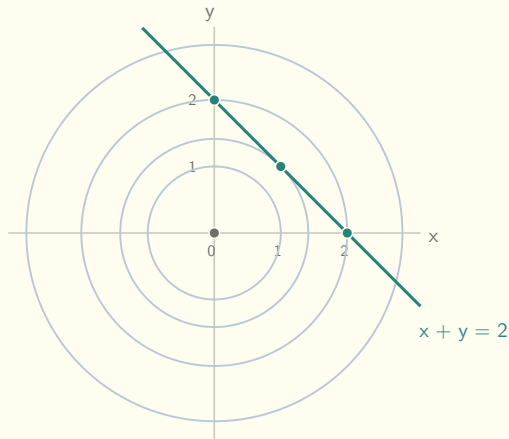
$$f(x, y) = x^2 + y^2$$

Each circle has one score.
Smaller circles have lower scores.

Without the constraint,
 $(0, 0)$ has the lowest score: 0.

Equality example: the feasible line

Only points on $x + y = 2$ are allowed



$$x + y = 2$$

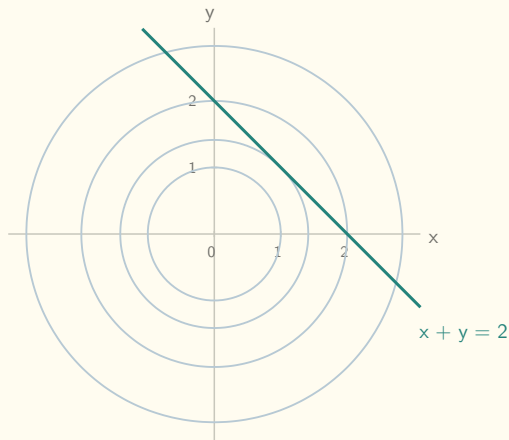
$$h(x, y) = 2 - x - y = 0$$

We may move along the line.
We may not leave it.

The origin is not allowed.

Recipe step 1: the Lagrangian

Combine the objective with the equality residual



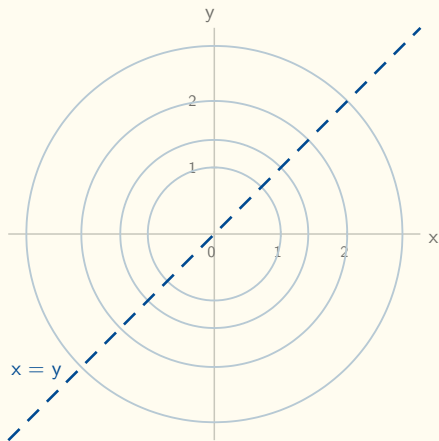
$$L = f + \lambda h$$

$$L = x^2 + y^2 + \lambda(2 - x - y)$$

On the feasible line:
 $h = 0$, so $L = f$.

Recipe step 2: stationarity

Both decision-variable derivatives must be zero



$$2x - \lambda = 0$$

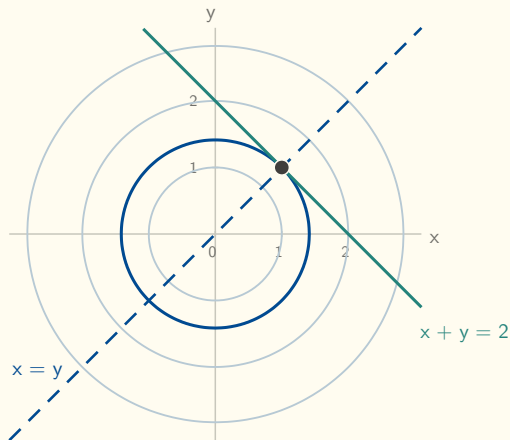
$$2y - \lambda = 0$$

Therefore: $x = y = \lambda/2$

The dashed line shows $x = y$.
Feasibility still needs checking.

Recipe step 2: feasibility

The candidate must also satisfy $x + y = 2$



$$x = y$$

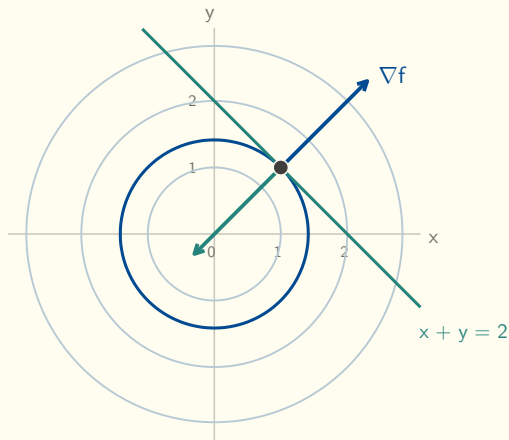
$$x + y = 2$$

$$x = 1, \quad y = 1$$

$$\lambda = 2x = 2$$

Recipe step 2: gradient balance

At the candidate, $\lambda = 2$ makes the vectors cancel



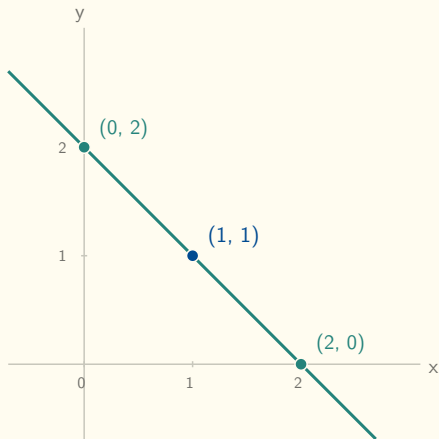
$$\nabla f = (2, 2)$$

$$\begin{aligned}\lambda \nabla h &= 2(-1, -1) \\ &= (-2, -2)\end{aligned}$$

$$\nabla f + \lambda \nabla h = 0$$

Verification: describe every allowed point

The equality $x + y = 2$ determines y once we choose x



$$x + y = 2$$

$$y = 2 - x$$

Every feasible point is $(x, 2 - x)$.

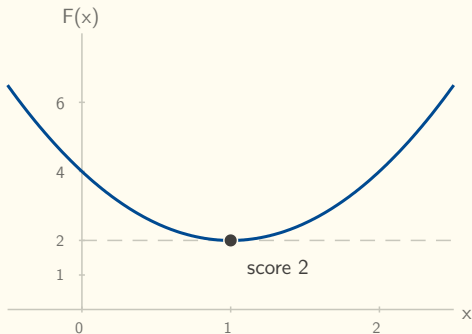
Its score is $f(x, 2 - x)$.

We are checking the whole line, not just the candidate.

Verification: no feasible point does better

Now compare the scores of every point $(x, 2 - x)$

Score along the line: $F(x)$



$$F(x) = x^2 + (2 - x)^2$$

$$= 2(x - 1)^2 + 2$$

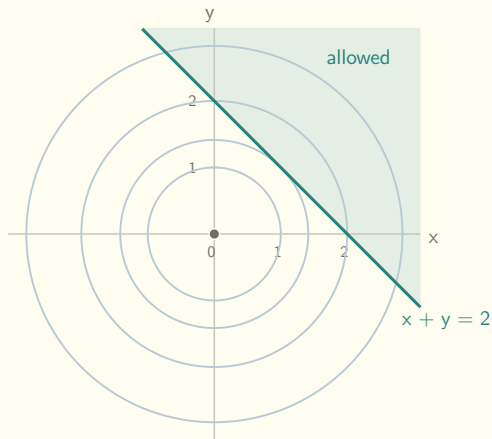
A square is always ≥ 0 .

So every feasible score is ≥ 2 .

$(1, 1)$ achieves 2. It is the minimum.

Inequality example: the feasible region

Minimize $x^2 + y^2$ subject to $x + y \geq 2$



minimize $x^2 + y^2$

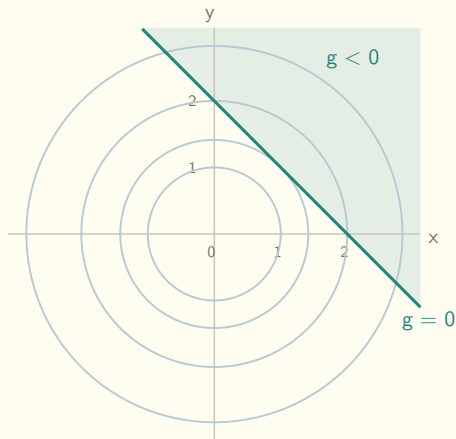
subject to $x + y \geq 2$

Which contour first reaches the shaded region?

Mark your predicted optimum.

Inequality example: the Lagrangian

Write the inequality in the form $g \leq 0$



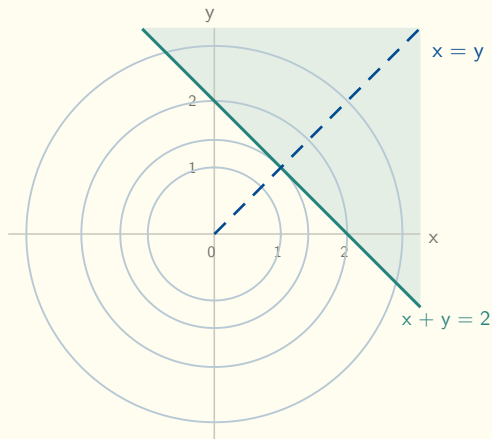
$$g(x, y) = 2 - x - y \leq 0$$

$$L = x^2 + y^2 \\ + \mu(2 - x - y)$$

$$\mu \geq 0$$

Inequality example: stationarity

Set the x and y slopes of L to zero



$$\partial L / \partial x = 2x - \mu = 0$$

$$\partial L / \partial y = 2y - \mu = 0$$

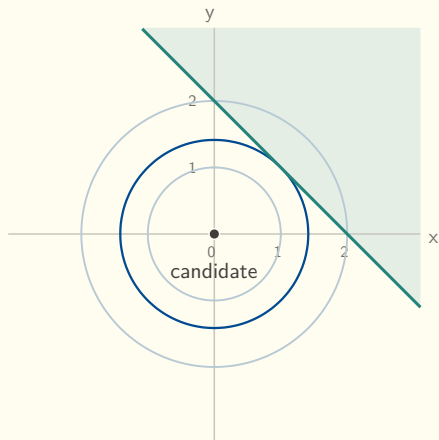
$$x = y = \mu/2$$

$$\text{Also: } \mu(2 - x - y) = 0.$$

Try $\mu = 0$, then try $g = 0$.

Inequality example: try $\mu = 0$

Minimize $x^2 + y^2$ subject to $x + y \geq 2$



Stationarity:

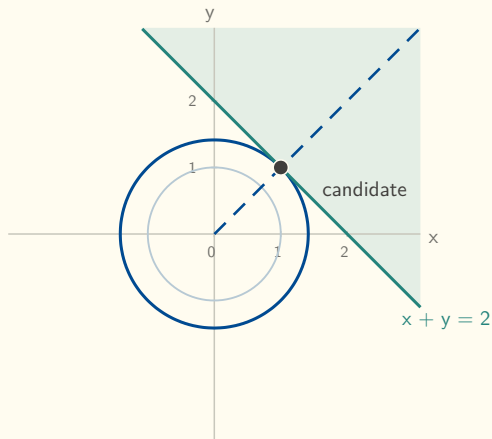
$$2x - \mu = 0, \quad 2y - \mu = 0$$

If $\mu = 0$: $x = y = 0$

The candidate lies outside the feasible region.

Inequality example: the active case

After $\mu = 0$ fails, try the boundary $g = 0$



$$x = y = \mu/2$$

$$x + y = 2$$

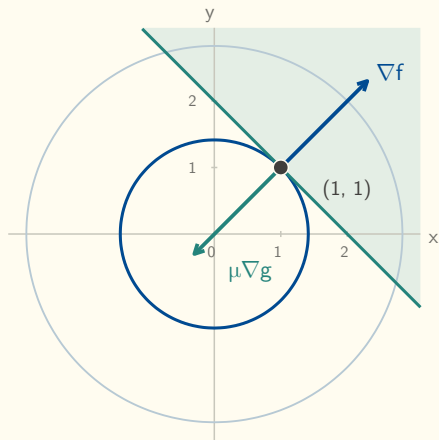
$$\mu/2 + \mu/2 = 2$$

$$\mu = 2$$

$$x = 1, \quad y = 1$$

Inequality example: all four checks

At $(1, 1)$, $\mu = 2$ satisfies the KKT conditions



Feasible: $g = 0$

Balanced: $\nabla f + \mu \nabla g = 0$

Sign: $\mu = 2 \geq 0$

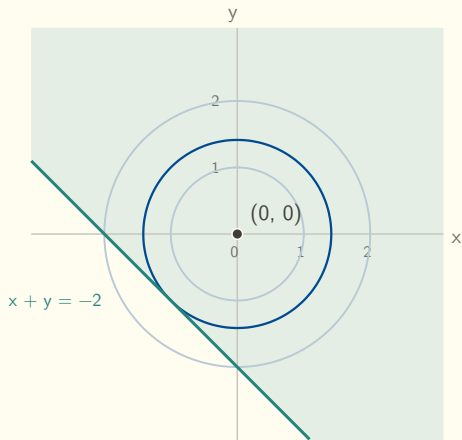
Slackness: $\mu g = 0$

Convexity proves the global minimum.

$$f(1, 1) = 2$$

Inequality example: room to spare

Minimize $x^2 + y^2$ subject to $x + y \geq -2$



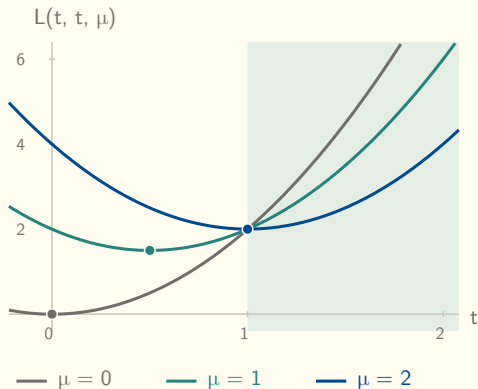
At (0, 0): $g = -2 < 0$

$$\nabla f = 0 \quad \mu^* = 0$$

The unconstrained optimum is already allowed.

First: minimize over the point

Each μ produces a bowl with its own minimum



$$2x - \mu = 0$$

$$2y - \mu = 0$$

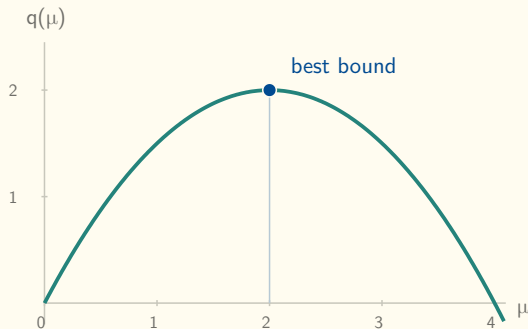
$$x = y = \mu/2$$

$$q(\mu) = 2\mu - \mu^2/2$$

Slice $x = y = t$. Dots mark the minima.

Then: maximize the lower bound

Here the dual function is a concave parabola



$$q(\mu) = 2\mu - \mu^2/2$$

$$q'(\mu) = 2 - \mu = 0$$

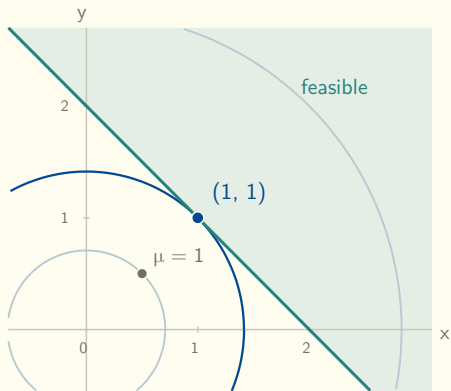
$$\mu^* = 2$$

$$q(2) = 2$$

Each height is a lower bound on the primal minimum.

A point and a bound meet

Together they certify the global optimum



Feasible point

$$f(1, 1) = 2$$

Dual lower bound

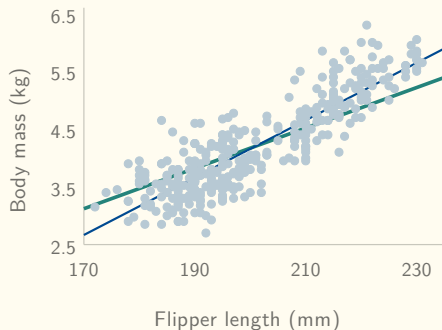
$$q(2) = 2$$

$\mu = 1$: $q(1) = 1.5$, but its inner minimizer is infeasible.

$$2 = q(2) \leq p^* \leq f(1, 1) = 2$$

ML example: a limit on the slope

Minimize squared error J , with a bound on the slope a



$$g(a, b) = a^2 - R^2 \leq 0$$

$$\nabla J + \mu \nabla g = 0$$

$$a = R = 0.7, \quad \mu > 0$$

The constraint is active.
Convexity makes this the
global best permitted fit.

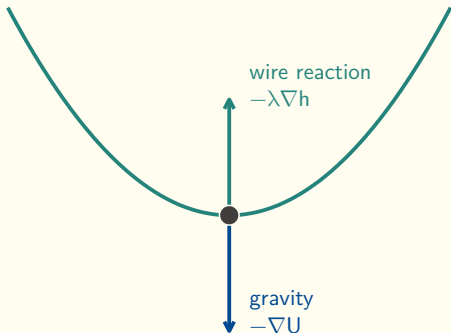
Palmer Penguins: 342 measured birds

— best without limit

— best permitted

Physics example: a bead on a wire

Minimize $U = mgz$ along a parabolic wire



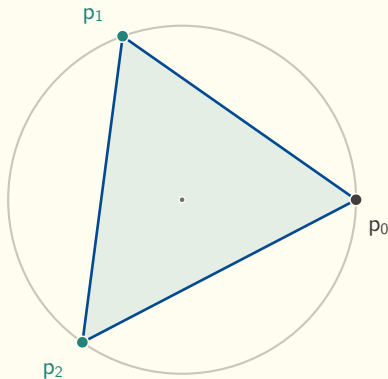
$$h = z - x^2/(2R) = 0$$

$$\nabla U + \lambda \nabla h = 0$$

The multiplier helps us find the force enforcing the constraint.

Your turn: the largest triangle

Two corners can move around a circle of radius 1



One corner is fixed.

$$p_0 = (1, 0)$$

maximize area A

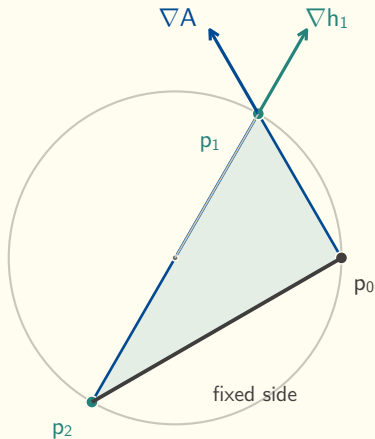
$$\|p_1\|^2 = 1$$

$$\|p_2\|^2 = 1$$

Where would you put the other two corners?

A hint: moving one corner

Hold the opposite side fixed



$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

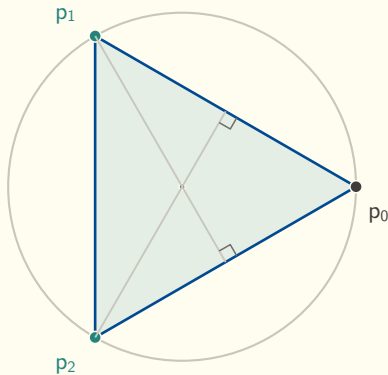
The area gradient is perpendicular to the opposite side.

The constraint gradient points away from the circle's centre.

At the optimum, these directions must align.

The triangle: equal sides

The multiplier condition must hold for both movable corners



Each radius is perpendicular to the opposite side.

$$p_1 \cdot p_2 = p_1 \cdot p_0$$

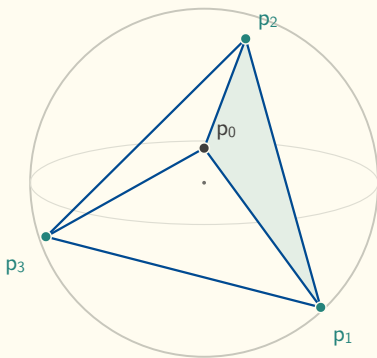
$$p_2 \cdot p_1 = p_2 \cdot p_0$$

All three sides have equal length.

Can the same argument work in every dimension?

Your turn: the largest simplex

A triangle in 2D, a tetrahedron in 3D, a simplex in n D



A tetrahedron, shown in 3D

Fix one vertex on the unit sphere.

Choose $p_1, \dots, p_n$ in $\mathbb{R}^n$.

maximize volume V

$$h_i = \|p_i\|^2 - 1 = 0$$

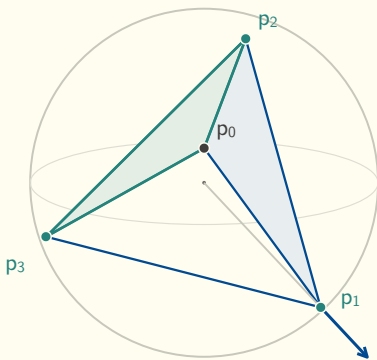
$$i = 1, \dots, n$$

Exactly n equality constraints.

n^2 scalar coordinates to choose.

The proof: two normal directions

At a maximum, the two normals must align for every free vertex



Green: the face opposite vertex 1

$$\nabla_i V = 2\lambda_i p_i$$

The volume gradient is normal to the opposite face.

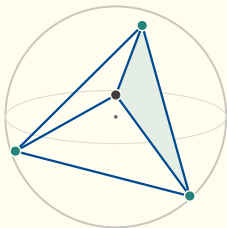
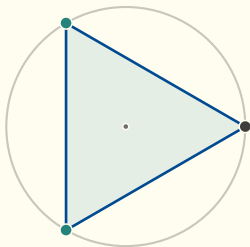
That face contains the edge from the fixed corner to each other corner.

$$p_i \cdot (p_j - p_0) = 0$$

for free vertices $i \neq j$

The result: a regular simplex

All pairs of vertices are equally far apart



Swapping i and j gives the same dot product.

$$p_i \cdot p_j = c \quad (i \neq j)$$

$$\|p_i - p_j\|^2 = 2 - 2c$$

All edges have equal length.
The maximizing simplex is regular.

A maximum exists, and the argument forces every maximum to have this shape.